
Traffic Accident Severity

Project Id: AIML-5

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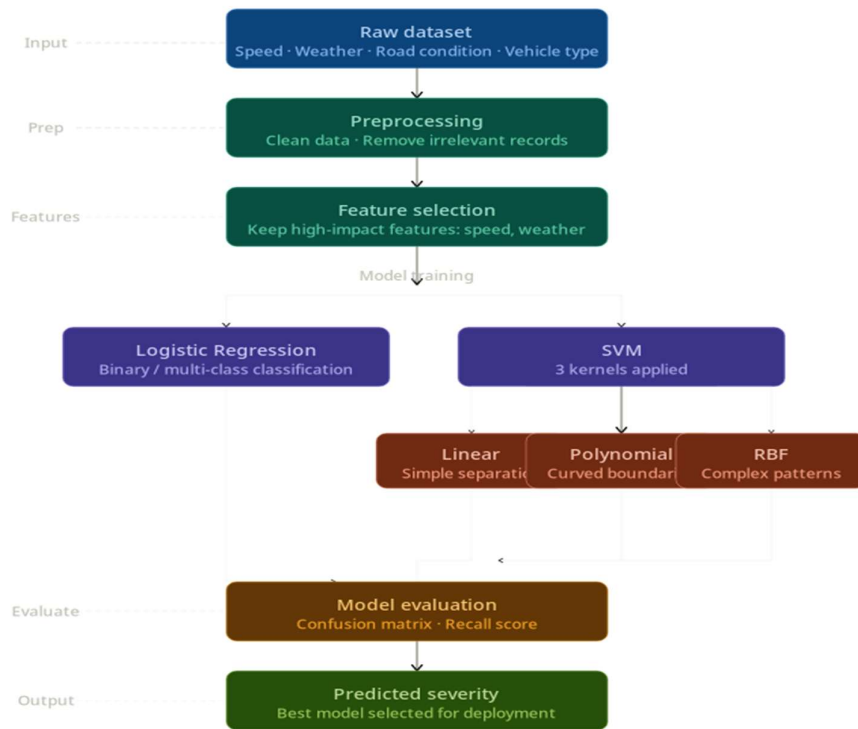
1. Problem Description

Road traffic accidents are a major global concern, causing significant loss of life, injuries, and property damage every year. Predicting the severity of accidents is an important task that can help authorities, emergency services, and transportation planners take preventive measures and improve road safety. However, accident severity depends on multiple factors such as vehicle speed, weather conditions, road conditions, and vehicle type, making it difficult to determine the potential impact of an accident using traditional analysis methods.

The aim of the Traffic Accident Severity Prediction system is to use machine learning techniques to analyze historical accident data and predict the severity level of future accidents. By processing and analyzing relevant features from accident datasets, the system can identify patterns and relationships between different factors that contribute to accident outcomes. Data preprocessing is performed to clean the dataset by removing irrelevant or missing records, ensuring better model performance. Feature selection is used to identify the most influential variables affecting accident severity.

Machine learning algorithms such as Logistic Regression and Support Vector Machine (SVM) with different kernels are applied to build predictive models. The performance of these models is evaluated using metrics such as confusion matrix and recall score. This predictive approach can help in improving traffic management, enhancing road safety strategies, and supporting faster emergency response systems.

2. Architecture flow diagram.



3. Models Used(Applied/ Planned) Give a brief introduction of the models used

The models used in this project are Support Vector Machine (SVM) and Logistic Regression. These are supervised machine learning algorithms commonly used for classification tasks and were selected for their accuracy and reliability.

Support Vector Machine (SVM)

SVM is used to classify data by finding the optimal boundary, called a hyperplane, that separates classes with maximum margin. Different kernels were applied in this project:

- **Linear SVM:** Suitable for linearly separable data using a straight-line boundary.
- **Polynomial SVM:** Handles non-linear data by creating curved decision boundaries.
- **RBF (Radial Basis Function) SVM:** Effective for complex datasets with highly non-linear patterns by mapping data into higher dimensions.
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Logistic Regression

Logistic Regression is a supervised learning algorithm used for binary and multi-class classification problems. It predicts the probability of an input belonging to a particular class using the sigmoid function. It is simple, efficient, and widely used for classification tasks.

Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve classification accuracy and reduce overfitting. It works by taking predictions from many trees and selecting the final result through majority voting. It is robust and performs well on large datasets.

Grid Search CV

Grid Search CV is a hyperparameter tuning technique used to find the best combination of model parameters. It uses cross-validation to test multiple parameter values and selects the combination that gives the highest performance. It helps in optimizing models like SVM, Logistic Regression, and Random Forest.

4.Evaluation metrics used

To measure the performance of the machine learning models, different evaluation metrics were used. These metrics help in understanding the accuracy and effectiveness of the models.

- **Accuracy Score:** Measures the overall percentage of correctly predicted instances out of total predictions. It is one of the most common metrics for classification tasks.
- **Precision:** Measures how many of the predicted positive cases were actually positive. It is important when false positives need to be minimized.
- **Recall:** Measures how many actual positive cases were correctly identified by the model. It is useful when false negatives need to be reduced.
- **F1-Score:** The harmonic mean of precision and recall. It provides a balanced measure when both false positives and false negatives are important.
- **Confusion Matrix:** A table used to visualize model performance by showing correct and incorrect predictions in terms of True Positive, True Negative, False Positive, and False Negative values.

5. Preliminary Results and Visualizations

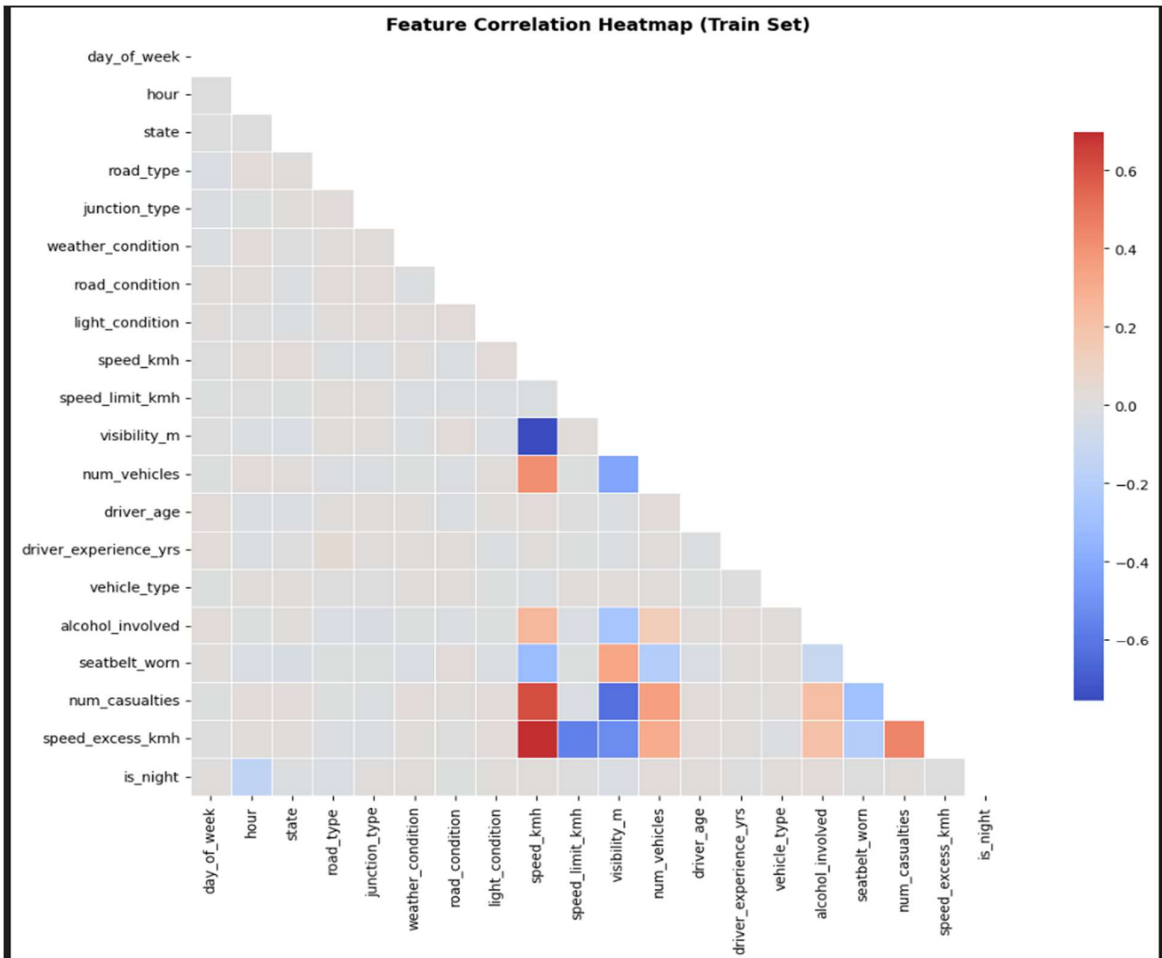


Fig.2 Heatmap showing correlation.

Logistic Regression

	precision	recall	f1-score	support
Minor	0.94	0.95	0.95	534
Moderate	0.91	0.91	0.91	533
Severe	0.97	0.96	0.97	533
accuracy			0.94	1600
macro avg	0.94	0.94	0.94	1600
weighted avg	0.94	0.94	0.94	1600

SVM Linear

	precision	recall	f1-score	support
Minor	0.95	0.95	0.95	534
Moderate	0.91	0.92	0.92	533
Severe	0.97	0.96	0.97	533
accuracy			0.94	1600
macro avg	0.94	0.94	0.94	1600
weighted avg	0.94	0.94	0.94	1600

SVM Polynomial

	precision	recall	f1-score	support
Minor	0.96	0.94	0.95	534
Moderate	0.88	0.94	0.91	533
Severe	0.98	0.93	0.96	533
accuracy			0.94	1600
macro avg	0.94	0.94	0.94	1600
weighted avg	0.94	0.94	0.94	1600

SVM RBF				

	precision	recall	f1-score	support
Minor	0.96	0.95	0.95	534
Moderate	0.91	0.93	0.92	533
Severe	0.98	0.96	0.97	533
accuracy			0.95	1600
macro avg	0.95	0.95	0.95	1600
weighted avg	0.95	0.95	0.95	1600

6. References:

Books:

Let Us Python – Yashavant Kanetkar

Online Resources:

Scikit-learn Documentation — <https://scikit-learn.org/>

Python Official Docs — <https://docs.python.org/>

GeeksforGeeks – <https://www.geeksforgeeks.org/>

GitHub – <https://github.com/>